# Angelica Knudsen

Allston, MA | P: 7013305705 | aknudsen@alum.mit.edu | angelicaknudsen.github.io

## SUMMARY

Recent MIT engineering graduate with 4+ years of hands-on experience in rapid prototyping, experimental R&D, and multi-platform CAD (SOLIDWORKS, Fusion 360, Onshape, AutoCAD). Experienced in leading student engineering teams and designing sustainable energy systems.

## EDUCATION

**Massachusetts Institute of Technology (MIT)** Cambridge, MA
Bachelor of Science, Mechanical Engineering with Energy May 2026
Cumulative GPA: 4.6/5.0

## EXPERIENCE

**Engineering Intern** Somerville, MA
Orien Energy Inc. Sep 2026 – Present

- Developing an adsorption heat pump to convert low-grade waste heat into valuable steam
- Conducting tests to characterize performance of the adsorbent bed, with LabJack as primary DAQ
- Processing and visualizing experiment results in MATLAB and Python
- Built the company website in HTML, CSS, and Javascript and wrote a website maintenance procedure

**Geo@MIT Researcher - Geothermal Heat Exchanger System Design** Cambridge, MA
MIT D-Lab Sep 2025 – Sep 2026

- Designed a pilot-scale heat exchanger system to investigate potential changes in water quality from using Cambridge water mains for a proposed thermal energy network at MIT
- Modeled assemblies of the mobile experiment (cart and water lines) containing 100+ components in SOLIDWORKS
- Built the cart and the portion of the experiment representing the ambient water loop system at MIT

**MIT Wind Build Lead** Cambridge, MA
MIT Edgerton Center Feb 2025 – May 2026

- Among 10 teams out of 30+ nationwide selected to compete in the 2025 and 2026 US Department of Energy's Collegiate Wind Competition
- Led the mechanical and electrical subteams, achieving a working prototype in 3 months
- Modeled system (tower and nacelle) in SOLIDWORKS and visualized performance curves in MATLAB
- Fabricated custom blades, designed to withstand 2800 RPM and 300N of centrifugal force

## PROJECTS

**Manta and Remora - Class Project for Engineering Systems Design** Sep 2025 – Dec 2025

- Co-designed a coupled remotely-operated vehicle (ROV) and unmanned surface vehicle (USV) to increase the cost-effectiveness, time-efficiency, and sustainability of offshore wind farm foundation inspections
- Contributed most to the custom reel mechanism, modeled in Onshape and designed to withstand over 1000N of force and hold 300m of tether for the ROV

## SKILLS

**Programming Languages**: Python, MATLAB, C++, C, LaTeX, HTML/CSS
**Manufacturing:** Laser Cutter, Mill, Bandsaw, Drill Press, Lathe, Shear, Brake, 3D printing, Soldering
**CAD:** SOLIDWORKS, Onshape, Autodesk Fusion 360, Rhino, AutoCAD